

Prediction Of Flares In RA Patient

Leveraging Machine Learning
on Flow Cytometry Data





Introduction

Overview of Rheumatoid Arthritis (RA)

- Chronic autoimmune disorder
- Characterized by inflammation, pain, and joint damage
- Relapsing and remitting nature

Hypothesis

- RA comprises distinct pathotypes with varied flare mechanisms

Challenges in Understanding RA Flares

- Mechanisms remain elusive despite advances
- Unpredictable nature complicates treatment planning

Importance of Personalized Treatment Strategies

- Tailoring treatments to individual patients
- Enhancing efficacy and minimizing side effects





Background

Overview of Bioflare Project

- 6-month observational study
- Collaboration between Newcastle University, University of Birmingham, and University of Glasgow



Research Focus

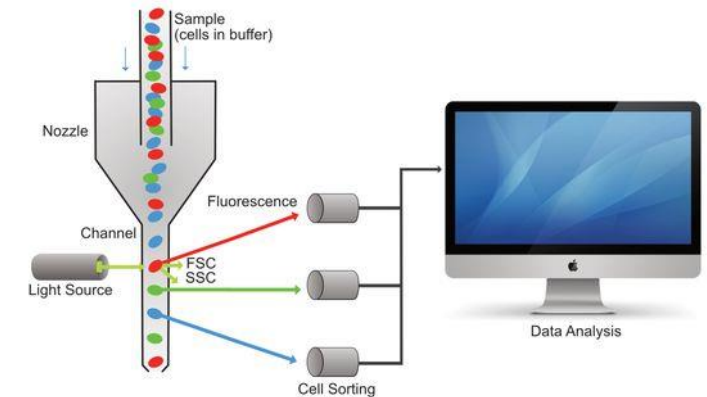
Addressing the lack of understanding in the pathogenesis of RA flares

Utilizing Advanced Techniques

Emphasis on flow cytometry for studying immune cell populations

Challenges and Solutions

- Manual analysis limitations
- Introduction of automated methods like OPTICS for improved analysis efficiency



Flow Cytometry

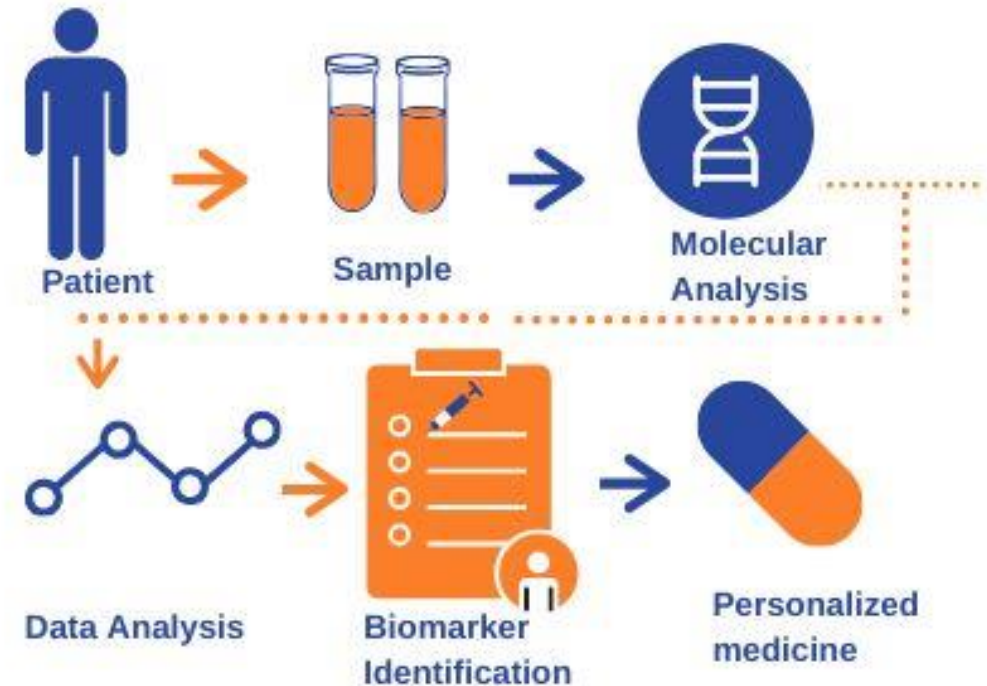


Research Question

Which subsets of biomarkers predict RA flares?

Tailoring treatments with biomarker subsets enhances efficacy, controls disease, and minimizes side effects, while gaining flare pathogenesis insights

Role of Biomarker





Objectives



Assessing Immune Cell Populations

- Measure the similarity or dissimilarity between immune cell populations
- Utilize distance metrics to quantify differences in immune cell profiles.



Identifying Distinct Subsets of RA Patients

- Utilizing Clustering to Distinguish Flare and Non-Flare RA Groups
- Examining Immune Cell Variations for RA Subgroup Identification

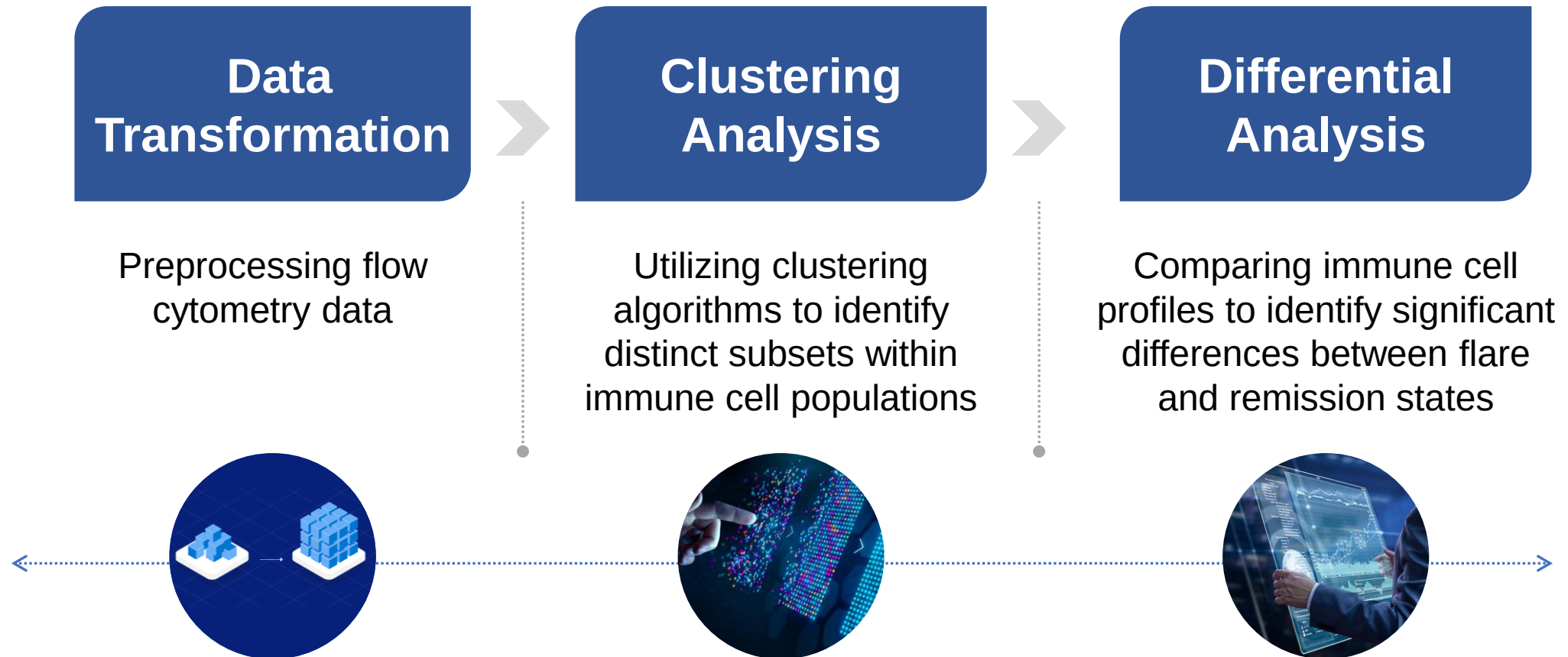


Comparing Immune Cell Profiles between Flare and Remission States

- Compare immune profiles of RA patients in flare vs. remission
- Identify significant biomarkers between RA flare and remission

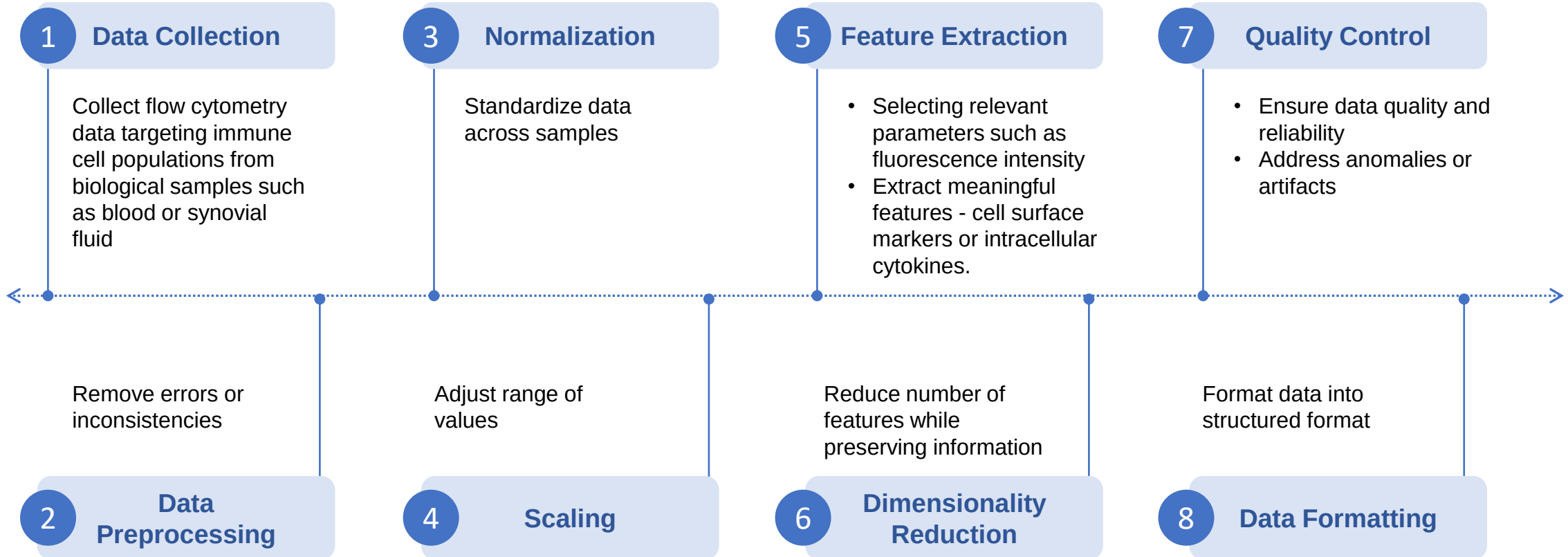


Methodology



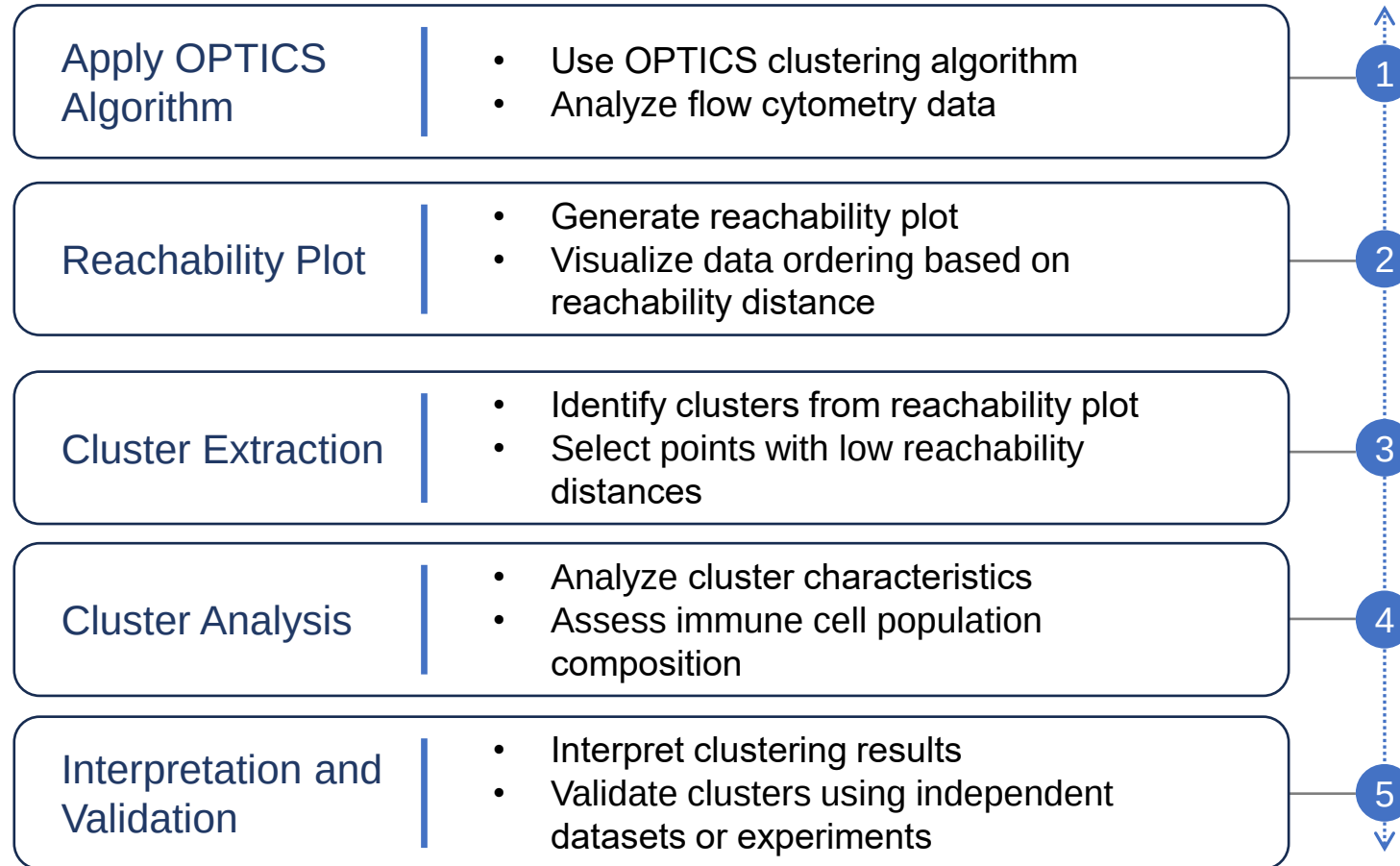


Data Transformation



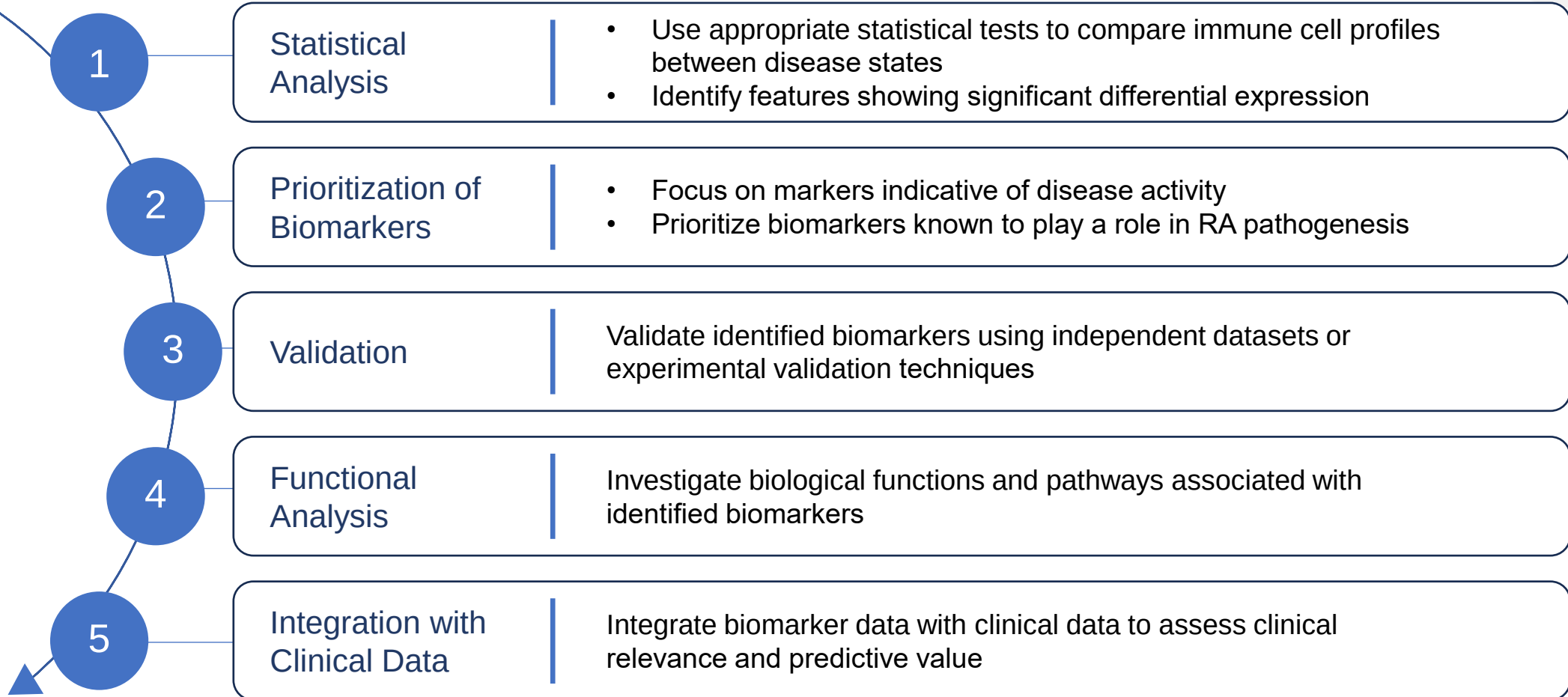


Clustering Analysis



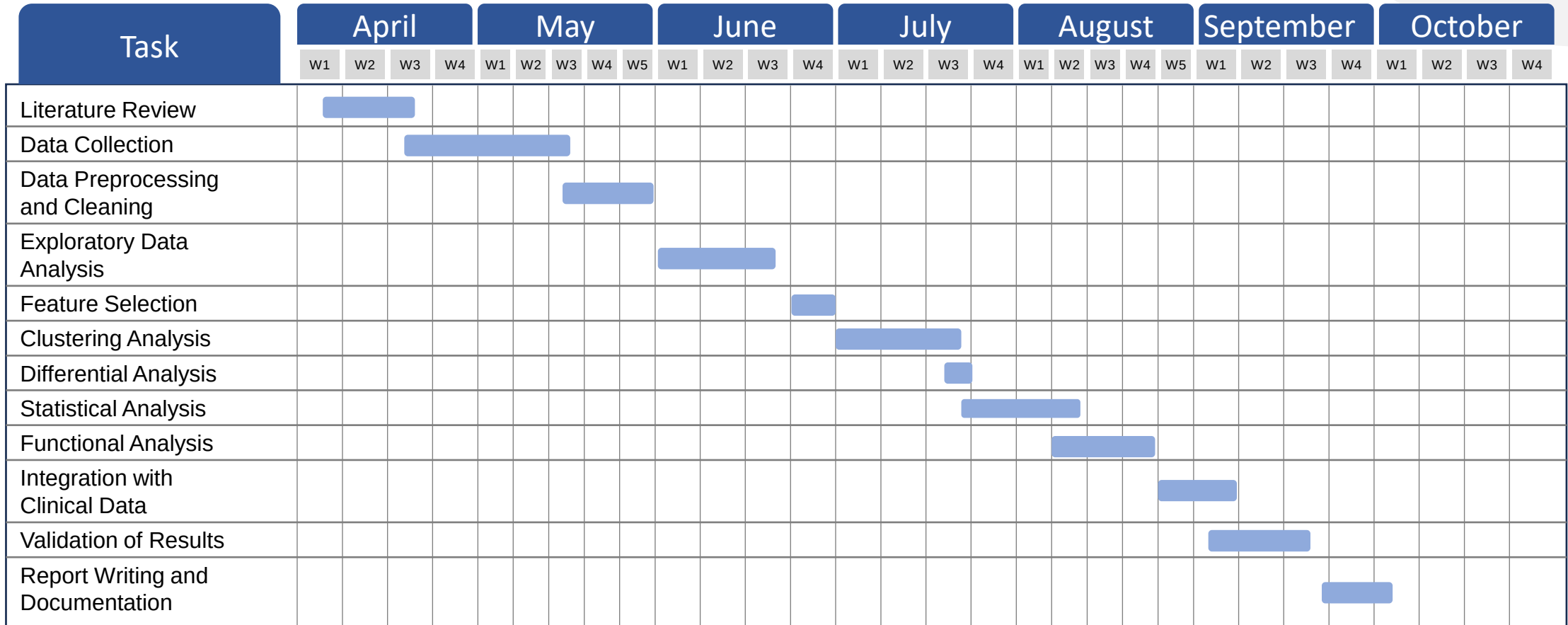


Differential Analysis





Timeline



Expected results:

- Prediction models for RA flares based on identified biomarkers
- Personalized treatment approaches



Summary

- Collaborative effort to understand and predict rheumatoid arthritis (RA) flares.
- Advanced methodologies like Optics and clustering analysis employed for data analysis.
- Focus on identifying predictive biomarkers for RA flare prediction.
- Expected outcomes include development of prediction models for RA flares and personalized treatment strategies.
- Detailed timeline outlines project phases from literature review to result validation.
- Future directions may involve further validation studies and clinical applications to improve RA management.